

CIRCLE EQUATIONS

Full Solutions & Explanations

DOCUMENT 2 — SOLUTIONS & EXPLANATIONS

Total Questions: 40

Q1. Correct Answer: **A • EASY**

STEP-BY-STEP SOLUTION:

The standard form of a circle's equation is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius.

Rewrite the given equation so both terms show subtraction: $(x - 6)^2 + (y - (-5))^2 = 64$.

Matching this to the standard form gives $h = 6$ and $k = -5$, so the center is $(6, -5)$.

WHY EACH OPTION IS WRONG:

- A)** Correct. $h = 6$ (since $x - 6$ matches $x - h$) and $k = -5$ (since $y + 5$ must be rewritten as $y - (-5)$).
- B)** This reads the -5 in the equation as if it were $+5$, ignoring that $(y + 5)^2$ means $k = -5$, not 5 .
- C)** This flips the sign on the x -coordinate while also misreading the y -coordinate as positive.
- D)** This flips the sign of h , treating $(x - 6)$ as if it meant $h = -6$.

TRAP ALERT

Seeing "+5" inside the parentheses and assuming the center's y -coordinate is positive 5 , instead of recognizing that the standard form always subtracts, so $+5$ really means $k = -5$.

KEY INSIGHT

In $(x - h)^2 + (y - k)^2 = r^2$, the center coordinates are always the **OPPOSITE** of the sign shown inside each parenthesis.

FAST METHOD

Flip the sign of whatever constant sits next to x and next to y — that flipped value is the center coordinate. No other work needed.

Q2. Correct Answer: **B • EASY**

STEP-BY-STEP SOLUTION:

In standard form $(x - h)^2 + (y - k)^2 = r^2$, the number on the right side of the equation is r^2 , not r . Here, $r^2 = 121$.

Take the square root of both sides: $r = \sqrt{121} = 11$.

WHY EACH OPTION IS WRONG:

- A)** This is r^2 , the value on the right side of the equation — the question asks for r , which requires taking the square root.
- B)** Correct. $r = \sqrt{121} = 11$.
- C)** This comes from dividing 121 by 2 instead of taking its square root.
- D)** This doubles the correct radius, possibly from confusing radius with diameter in the wrong direction.

TRAP ALERT

Stopping as soon as you see the number on the right-hand side and reporting it directly as the radius, without remembering that number is r^2 .

KEY INSIGHT 💡

The constant on the right side of a circle's standard-form equation is always r^2 . You must take its square root to get the actual radius.

FAST METHOD ⚡

Recognize perfect squares on sight: $121 = 11^2$, so $r = 11$ immediately.

Q3. Correct Answer: **C • EASY**

STEP-BY-STEP SOLUTION:

Start from the standard form $(x - h)^2 + (y - k)^2 = r^2$.

Substitute the center $(h, k) = (-3, 8)$: $x - (-3)$ becomes $(x + 3)$, and $y - 8$ stays $(y - 8)$.

Substitute $r = 5$, so $r^2 = 25$.

The equation is $(x + 3)^2 + (y - 8)^2 = 25$.

WHY EACH OPTION IS WRONG:

- A)** This uses $r = 5$ directly on the right side instead of squaring it to get $r^2 = 25$.
- B)** This flips the signs on both coordinates, treating the center as if it were $(3, -8)$ instead of $(-3, 8)$.
- C)** Correct. Center $(-3, 8)$ gives $(x + 3)^2 + (y - 8)^2$, and radius 5 gives $r^2 = 25$.
- D)** This doubles the radius ($5 \times 2 = 10$) instead of squaring it ($5^2 = 25$).

TRAP ALERT 🚧

After correctly building the $(x - h)$ and $(y - k)$ parts, forgetting that the radius must be squared before it goes on the right-hand side.

KEY INSIGHT 💡

Building a circle equation is a two-part process: flip the sign of each center coordinate for the parentheses, and square the radius for the right side.

FAST METHOD ⚡

Write h and k with flipped signs inside the parentheses, then just square r mentally before writing the final number.

Q4. Correct Answer: **D • EASY**

STEP-BY-STEP SOLUTION:

The standard form is $(x - h)^2 + (y - k)^2 = r^2$. The given equation has x^2 with no visible subtraction, but this means $x^2 = (x - 0)^2$, so $h = 0$.

Rewrite $(y + 4)^2$ as $(y - (-4))^2$, so $k = -4$.

The center is $(0, -4)$.

WHY EACH OPTION IS WRONG:

- A)** This correctly identifies $h = 0$ but misreads the sign on k , treating $+4$ as positive instead of recognizing $k = -4$.
- B)** This swaps which coordinate is zero, placing the missing term on y instead of x .
- C)** This swaps the coordinates entirely and also gets the sign wrong on the nonzero coordinate.

D) Correct. x^2 means $h = 0$, and $(y + 4)^2$ means $k = -4$.

TRAP ALERT 🚧

When a term like x^2 appears with no parentheses or subtraction sign, it's easy to forget that it still represents $(x - 0)^2$, so the corresponding center coordinate is exactly 0, not skipped or guessed.

KEY INSIGHT 💡

A bare x^2 or y^2 term always means that coordinate of the center is 0 — the circle is centered on an axis.

FAST METHOD ⚡

If a variable appears squared with nothing subtracted from it, its center coordinate is automatically 0.

Q5. Correct Answer: **6** • EASY 🖋️ GRID-IN

STEP-BY-STEP SOLUTION:

The radius is the distance from the center to any point on the circle.

Since the center (2, 3) and the point (2, 9) share the same x-coordinate, the distance between them is just the difference in y-coordinates: $|9 - 3| = 6$.

So the radius is 6.

COMMON WRONG ANSWERS & WHY:

- Gridding 36 — this is r^2 (6^2), not r itself. A student who builds the equation $(x-2)^2 + (y-3)^2 = 36$ may stop there instead of answering the radius question.
- Gridding 9 or 3 — entering one of the given coordinates instead of computing the actual distance between the two points.

TRAP ALERT 🚧

Because the points share an x-coordinate, students sometimes skip the distance formula but then mistakenly report one of the y-coordinates itself rather than their difference.

KEY INSIGHT 💡

The radius is the distance from center to any point on the circle — always a positive length, found using the distance formula or, when coordinates align, a simple subtraction.

FAST METHOD ⚡

When two points share an x- or y-coordinate, the distance between them is just the absolute difference of the other coordinate — no formula needed.

Q6. Correct Answer: **A** • EASY

STEP-BY-STEP SOLUTION:

From the standard form, $r^2 = 81$, so $r = \sqrt{81} = 9$.

The diameter is twice the radius: $d = 2r = 2(9) = 18$.

WHY EACH OPTION IS WRONG:

- A)** Correct. $r = 9$, so the diameter is $2(9) = 18$.
- B)** This is the radius, not the diameter — the question asks for twice this value.
- C)** This is r^2 , the constant from the equation, not a length measurement at all.
- D)** This comes from dividing $r^2 = 81$ by 2, instead of first taking the square root and then doubling.

TRAP ALERT

Stopping after finding the radius and forgetting the question asked for the diameter, which requires one more step.

KEY INSIGHT

Diameter = $2 \times$ radius. Always check which length the question is actually asking for.

FAST METHOD

Recognize $81 = 9^2$ instantly, then double: $9 \times 2 = 18$.

Q7. Correct Answer: **B • EASY****STEP-BY-STEP SOLUTION:**

From the standard form, $r^2 = 49$, so $r = 7$.

Circumference is given by $C = 2\pi r$.

$$C = 2\pi(7) = 14\pi.$$

WHY EACH OPTION IS WRONG:

- A)** This uses $r = 7$ directly without multiplying by 2, omitting half of the formula.
- B)** Correct. $C = 2\pi r = 2\pi(7) = 14\pi$.
- C)** This applies the area formula $\pi r^2 = \pi(49)$ instead of the circumference formula $2\pi r$.
- D)** This treats the diameter (14) as if it were the radius, then multiplies by 2π again, doubling the answer.

TRAP ALERT

Confusing the circumference formula ($2\pi r$) with the area formula (πr^2), especially when r^2 is the number sitting right there in the equation.

KEY INSIGHT

Circumference uses r to the first power ($2\pi r$); area uses r squared (πr^2). Keep these two formulas clearly separated.

FAST METHOD

Take $\sqrt{49} = 7$ first, then just double it before attaching π : $7 \times 2 = 14$, so 14π .

Q8. Correct Answer: **C • EASY****STEP-BY-STEP SOLUTION:**

Rewrite $(x + 10)^2$ as $(x - (-10))^2$ to match the standard form $(x - h)^2 + (y - k)^2 = r^2$.

This gives $h = -10$. The term $(y - 3)^2$ directly gives $k = 3$.

The center is $(-10, 3)$.

WHY EACH OPTION IS WRONG:

- A)** This flips the sign on the x-coordinate, treating +10 inside the parentheses as if it meant $h = 10$.
- B)** This flips the sign on the x-coordinate only, getting y right but x wrong.
- C)** Correct. $h = -10$ (because $x + 10$ means $x - (-10)$) and $k = 3$.
- D)** This flips the sign on the y-coordinate while leaving x correct, the opposite mistake of choice B.

TRAP ALERT

Misjudging which coordinate's sign needs flipping when the x-term and y-term show different signs (one addition, one subtraction) in the same equation.

KEY INSIGHT 💡

Apply the sign-flip rule independently to each coordinate — don't let the sign pattern of one term influence how you read the other.

FAST METHOD ⚡

Flip the x-term's sign and flip the y-term's sign separately and immediately; never assume they match.

Q9. Correct Answer: **169** • EASY 🖊️ GRID-IN

STEP-BY-STEP SOLUTION:

Since the circle is centered at the origin, its equation has the form $x^2 + y^2 = r^2$, so $c = r^2$.

The radius equals the distance from the origin (0, 0) to the point (5, 12) on the circle.

$r^2 = 5^2 + 12^2 = 25 + 144 = 169$, so $c = 169$.

COMMON WRONG ANSWERS & WHY:

- Gridding 13 — this is r itself ($\sqrt{169} = 13$), but the question asks for c , which equals r^2 , not r .
- Gridding 17 — this comes from adding $5 + 12$ directly instead of using the Pythagorean relationship (squaring and adding).

TRAP ALERT 🚧

Solving all the way down to the radius (13) and gridding that, instead of noticing the question defines c as r^2 , not r .

KEY INSIGHT 💡

For a circle centered at the origin, $x^2 + y^2 = r^2$ — any point (x, y) on the circle satisfies this directly, no need to find r first if only r^2 (i.e., c) is needed.

FAST METHOD ⚡

Just square and add the coordinates of the given point: $5^2 + 12^2 = 169 = c$, skipping the square root entirely.

Q10. Correct Answer: **D** • EASY

STEP-BY-STEP SOLUTION:

From the standard form, $r^2 = 100$ directly.

The area of a circle is $A = \pi r^2$.

$A = \pi(100) = 100\pi$.

WHY EACH OPTION IS WRONG:

- A)** This takes $\sqrt{100} = 10$ first and multiplies by π , applying the circumference-style thinking (using r , not r^2) to the area formula.
- B)** This uses the diameter (20) multiplied by π instead of r^2 multiplied by π .
- C)** This halves the correct area, possibly from miscombining r and r^2 .
- D)** Correct. Since $A = \pi r^2$ and $r^2 = 100$ is already given directly in the equation, $A = 100\pi$.

TRAP ALERT 🚧

Taking the unnecessary extra step of finding $r = \sqrt{100} = 10$ first, when the area formula πr^2 actually needs r^2 , which was already sitting in the equation.

KEY INSIGHT

Because area $= \pi r^2$, and a circle's equation already gives you r^2 directly, you can find the area in one step without ever computing r .

FAST METHOD

Skip finding r entirely — just multiply the right-hand side of the equation by π .

Q11. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

Group x-terms and y-terms, then move the constant to the right side: $(x^2 - 6x) + (y^2 + 8y) = 11$.

Complete the square for x: $x^2 - 6x$ becomes $(x - 3)^2 - 9$. Complete the square for y: $y^2 + 8y$ becomes $(y + 4)^2 - 16$.

Substitute back: $(x - 3)^2 - 9 + (y + 4)^2 - 16 = 11$.

Add 9 and 16 to both sides: $(x - 3)^2 + (y + 4)^2 = 11 + 9 + 16 = 36$.

So $r^2 = 36$, which means $r = 6$.

WHY EACH OPTION IS WRONG:

A) Correct. After completing the square, $r^2 = 36$, so $r = \sqrt{36} = 6$.

B) This is r^2 , not r — the square root step was skipped.

C) This uses the original constant term (-11 , reported as 11) directly instead of completing the square first.

D) This adds only the two completing-the-square constants ($9 + 16 = 25$, $\sqrt{25} = 5$) but forgets to also incorporate the original -11 constant term correctly.

TRAP ALERT

Forgetting to move the constant term to the other side of the equation before completing the square, which throws off the final value under the square root.

KEY INSIGHT

To find the radius from general form, always isolate the constant first, complete the square on x and y separately, then add everything completed to both sides before reading off r^2 .

FAST METHOD

Half the x-coefficient and square it, half the y-coefficient and square it, add both results plus whatever was isolated on the other side — that total is r^2 .

Q12. Correct Answer: **B** • MEDIUM

STEP-BY-STEP SOLUTION:

Group and isolate the constant: $(x^2 + 10x) + (y^2 - 2y) = -13$.

Complete the square: $x^2 + 10x$ becomes $(x + 5)^2 - 25$. $y^2 - 2y$ becomes $(y - 1)^2 - 1$.

Substitute: $(x + 5)^2 - 25 + (y - 1)^2 - 1 = -13$, so $(x + 5)^2 + (y - 1)^2 = -13 + 25 + 1 = 13$.

Reading off the center from $(x + 5)^2 + (y - 1)^2$: $h = -5$ and $k = 1$, so the center is $(-5, 1)$.

WHY EACH OPTION IS WRONG:

A) This flips the sign on both coordinates, treating $(x + 5)$ as $h = 5$ and $(y - 1)$ as $k = -1$.

- B)** Correct. $(x + 5)^2$ gives $h = -5$, and $(y - 1)^2$ gives $k = 1$.
C) This correctly flips the x-coordinate's sign but incorrectly also flips the y-coordinate's sign.
D) This keeps the x-coordinate's sign as written (not flipping it) and also flips the y-coordinate incorrectly.

TRAP ALERT

After successfully completing the square, rushing the final sign-reading step and flipping (or failing to flip) the wrong coordinate.

KEY INSIGHT

Completing the square always produces a binomial like $(x + p)^2$, and the center coordinate is always $-p$, regardless of how the original equation was arranged.

FAST METHOD

Take half of the x-coefficient and negate it for h ; take half of the y-coefficient and negate it for k : $h = -10/2 = -5$, $k = -(-2)/2 = 1$.

Q13. Correct Answer: **6** • MEDIUM  GRID-IN

STEP-BY-STEP SOLUTION:

Isolate the constant: $(x^2 - 4x) + (y^2 - 12y) = -4$.

Complete the square: $x^2 - 4x$ becomes $(x - 2)^2 - 4$. $y^2 - 12y$ becomes $(y - 6)^2 - 36$.

Substitute: $(x - 2)^2 - 4 + (y - 6)^2 - 36 = -4$, so $(x - 2)^2 + (y - 6)^2 = -4 + 4 + 36 = 36$.

So $r^2 = 36$, which means $r = 6$.

COMMON WRONG ANSWERS & WHY:

- Gridding 36 — this is r^2 , the value right after completing the square. Forgetting to take the final square root is the single most common error in this question type.
- Gridding 4 — mistakenly using the original constant term from the equation instead of the value obtained after completing the square.

TRAP ALERT

Reaching the correct equation $(x - 2)^2 + (y - 6)^2 = 36$ and stopping there, reporting 36 as the radius instead of taking one more square root.

KEY INSIGHT

Completing the square always gets you to r^2 first — the radius itself requires one final square root.

FAST METHOD

Half of -4 is -2 , squared is 4. Half of -12 is -6 , squared is 36. Add those plus the isolated -4 on the other side: $4 + 36 - 4 = 36$, then $\sqrt{36} = 6$.

Q14. Correct Answer: **C** • MEDIUM

STEP-BY-STEP SOLUTION:

The center of a circle is the midpoint of any diameter.

Use the midpoint formula: $((x_1 + x_2)/2, (y_1 + y_2)/2)$ with $(-1, 4)$ and $(7, -2)$.

x-coordinate: $(-1 + 7)/2 = 6/2 = 3$. y-coordinate: $(4 + (-2))/2 = 2/2 = 1$.

The center is $(3, 1)$.

WHY EACH OPTION IS WRONG:

- A)** This adds the coordinates correctly ($-1 + 7 = 6$, $4 + (-2) = 2$) but forgets to divide each sum by 2.
- B)** This correctly finds the x-coordinate but makes a sign error on the y-coordinate, computing $4 - (-2)$ interpreted incorrectly.
- C)** Correct. The midpoint is $((-1+7)/2, (4+(-2))/2) = (3, 1)$.
- D)** This makes a sign error on the x-coordinate while getting y correct.

TRAP ALERT

Computing the sum of corresponding coordinates correctly but forgetting the essential last step of dividing by 2 to actually find the midpoint.

KEY INSIGHT

The center of a circle is always the midpoint of a diameter: average the x-coordinates and average the y-coordinates separately.

FAST METHOD

Add the x-values and divide by 2; add the y-values and divide by 2. Two quick calculations, no formula needed beyond basic averaging.

Q15. Correct Answer: **D** • MEDIUM

STEP-BY-STEP SOLUTION:

First find the length of the diameter using the distance formula between the two endpoints.

$$d = \sqrt{[(7 - (-1))]^2 + (-2 - 4)^2} = \sqrt{8^2 + (-6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10.$$

The radius is half the diameter: $r = 10/2 = 5$.

WHY EACH OPTION IS WRONG:

- A)** This is the diameter, not the radius — the final step of dividing by 2 was skipped.
- B)** This halves the squared distance ($100/2 = 50$) instead of halving the actual distance after taking the square root.
- C)** This uses only the horizontal leg of the right triangle formed by the two points ($\Delta x = 8$), ignoring the vertical leg entirely.
- D)** Correct. The distance between the endpoints is 10 (the diameter), so the radius is $10/2 = 5$.

TRAP ALERT

Correctly computing the distance between the two given points but forgetting that this distance is the full diameter, not the radius.

KEY INSIGHT

When a question gives diameter endpoints, the distance formula gives you the diameter first — always divide by 2 to get the radius.

FAST METHOD

Recognize the 6-8-10 right triangle pattern immediately (legs 8 and 6 give hypotenuse 10), then just halve it: $10/2 = 5$.

Q16. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

The center is the midpoint of the diameter: $(3, 1)$, found using the midpoint formula on $(-1, 4)$ and $(7, -2)$.

The diameter length is $\sqrt{8^2 + (-6)^2} = \sqrt{100} = 10$, so the radius is 5 and $r^2 = 25$.
Substituting into standard form: $(x - 3)^2 + (y - 1)^2 = 25$.

WHY EACH OPTION IS WRONG:

- A)** Correct. Center (3, 1) and $r^2 = 25$ give $(x - 3)^2 + (y - 1)^2 = 25$.
- B)** This flips the signs of both center coordinates, as if the center were $(-3, -1)$ instead of (3, 1).
- C)** This uses $d^2 = 100$ (the diameter squared) on the right side instead of $r^2 = 25$.
- D)** This uses $r = 5$ directly without squaring it to get $r^2 = 25$.

TRAP ALERT

After correctly finding diameter = 10, accidentally squaring the diameter (getting 100) instead of first halving it to get the radius and then squaring that.

KEY INSIGHT

Building the equation from diameter endpoints requires two separate calculations: the midpoint gives the center, and half the distance (then squared) gives r^2 .

FAST METHOD

Once you know the center and that the legs of the distance triangle were 6 and 8 (giving hypotenuse 10), recall $r = 5$ and $r^2 = 25$ directly.

Q17. Correct Answer: **-36** • MEDIUM  GRID-IN

STEP-BY-STEP SOLUTION:

Isolate the constant: $(x^2 + 6x) + (y^2 - 4y) = -c$.

Complete the square: $x^2 + 6x$ becomes $(x + 3)^2 - 9$. $y^2 - 4y$ becomes $(y - 2)^2 - 4$.

Substitute: $(x + 3)^2 - 9 + (y - 2)^2 - 4 = -c$, so $(x + 3)^2 + (y - 2)^2 = -c + 13$.

Since the radius is 7, $r^2 = 49$, so $-c + 13 = 49$.

Solve: $-c = 49 - 13 = 36$, so $c = -36$.

COMMON WRONG ANSWERS & WHY:

- Gridding 36 — forgetting the negative sign that results from solving $-c = 36$ for c .
- Gridding 62 — adding 49 and 13 instead of subtracting, from a sign error when isolating c .

TRAP ALERT

Losing track of the negative sign in front of c during the algebra, since c was originally written as $+c$ but ends up needing to be negative for the radius to equal 7.

KEY INSIGHT

When a variable like c sits inside the general-form equation, complete the square as usual, then set the resulting expression equal to the given r^2 and solve carefully for c , watching all signs.

FAST METHOD

The two completing-the-square constants always add to give the 'extra' amount ($9 + 4 = 13$ here). Set $13 - c = r^2$ and solve directly: $c = 13 - r^2 = 13 - 49 = -36$.

Q18. Correct Answer: **B** • MEDIUM

STEP-BY-STEP SOLUTION:

To compare point P to the circle, substitute P's coordinates into the left side of the equation and compare the result to $r^2 = 50$.

$$(10 - 4)^2 + (-1 + 1)^2 = 6^2 + 0^2 = 36 + 0 = 36.$$

Since $36 < 50$, point P is closer to the center than the radius allows, so P lies inside the circle.

WHY EACH OPTION IS WRONG:

- A)** This would require the substituted value to equal exactly 50, but it equals 36, so P is not on the circle.
- B)** Correct. Since $36 < 50$, the point is closer to the center than the radius, placing it inside the circle.
- C)** This would require the substituted value to exceed 50, but 36 is less than 50, so P is not outside.
- D)** P's location can absolutely be determined by substitution — this information is sufficient.

TRAP ALERT

Computing the substituted value correctly but then misremembering whether a smaller result means inside or outside the circle.

KEY INSIGHT

Substitute the point into $(x - h)^2 + (y - k)^2$ and compare to r^2 : a smaller result means inside, equal means on, and larger means outside the circle.

FAST METHOD

Compute the substituted value once, then just compare directly to the r^2 already given in the equation — no need to take any square roots.

Q19. Correct Answer: **C** • MEDIUM

STEP-BY-STEP SOLUTION:

From circle A's equation, $r^2 = 16$, so the radius of circle A is $\sqrt{16} = 4$.

Circle B has 3 times this radius: $3 \times 4 = 12$.

Since circle B shares the same center, only the right-hand side changes, using the new radius squared: $12^2 = 144$.

Circle B's equation is $(x + 6)^2 + (y - 2)^2 = 144$.

WHY EACH OPTION IS WRONG:

- A)** This multiplies $r^2 = 16$ by 3 directly ($16 \times 3 = 48$), but scaling the radius by 3 means r^2 scales by $3^2 = 9$, not by 3.
- B)** This correctly finds the new radius (12) but forgets to square it before placing it on the right side of the equation.
- C)** Correct. New radius = $3(4) = 12$, and $r^2 = 12^2 = 144$.
- D)** This adds 3 to the original $r^2 = 16$ (getting 19) instead of scaling it by the radius multiplier.

TRAP ALERT

Scaling r^2 by the same factor as r itself, forgetting that if the radius is multiplied by 3, the squared radius is multiplied by $3^2 = 9$, not by 3.

KEY INSIGHT

Whenever a radius is scaled by some factor k , the value of r^2 in the equation scales by k^2 , since r^2 is being squared, not just multiplied.

FAST METHOD

Find the new radius first (multiply the old radius, not r^2 , by the scale factor), then square that new radius to get the new right-hand side.

Q20. Correct Answer: **D** • MEDIUM

STEP-BY-STEP SOLUTION:

Isolate the constant: $(x^2 - 8x) + (y^2 + 2y) = 8$.

Complete the square: $x^2 - 8x$ becomes $(x - 4)^2 - 16$. $y^2 + 2y$ becomes $(y + 1)^2 - 1$.

Substitute: $(x - 4)^2 - 16 + (y + 1)^2 - 1 = 8$, so $(x - 4)^2 + (y + 1)^2 = 8 + 16 + 1 = 25$.

So $r^2 = 25$. The area is $A = \pi r^2 = 25\pi$.

WHY EACH OPTION IS WRONG:

A) This takes $\sqrt{25} = 5$ and multiplies by π , applying the circumference-style thinking (using r alone) to the area question.

B) This uses the diameter (10) multiplied by π instead of r^2 multiplied by π .

C) This doubles the correct area ($25\pi \times 2 = 50\pi$), possibly from confusing r^2 with something requiring an extra factor of 2.

D) Correct. After completing the square, $r^2 = 25$, so the area is $\pi(25) = 25\pi$.

TRAP ALERT 🚧

Successfully completing the square to find $r^2 = 25$, but then taking the unnecessary extra step of computing $r = 5$ before applying the area formula, sometimes leading to using r instead of r^2 .

KEY INSIGHT 💡

Once you've completed the square and found r^2 , the area formula $A = \pi r^2$ can be applied immediately — no need to ever compute r itself.

FAST METHOD ⚡

After completing the square, just attach π to the r^2 value you already found: 25 becomes 25π directly.

Q21. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

The boundary of the watered region is a circle centered at the sprinkler's location, (12, 9), with radius equal to the maximum reach, 15 feet.

Using standard form with $h = 12$, $k = 9$, and $r = 15$: $(x - 12)^2 + (y - 9)^2 = r^2$.

$r^2 = 15^2 = 225$, so the equation is $(x - 12)^2 + (y - 9)^2 = 225$.

WHY EACH OPTION IS WRONG:

A) Correct. Center (12, 9) and radius 15 give $(x - 12)^2 + (y - 9)^2 = 225$.

B) This places $r = 15$ directly on the right side instead of squaring it to get $r^2 = 225$.

C) This flips the signs of both coordinates, placing the center at $(-12, -9)$ instead of $(12, 9)$.

D) This swaps the x - and y -coordinates of the sprinkler's location.

TRAP ALERT 🚧

Reading the real-world numbers (12, 9, and 15) correctly but forgetting that the radius value (15) must still be squared before it becomes the constant in the equation.

KEY INSIGHT 💡

Word problems describing a circular boundary always translate directly into standard form: the location is the center, and the maximum distance is the radius.

FAST METHOD ⚡

Identify the location as (h, k) and the maximum distance as r , then immediately square r for the right-hand side: $15^2 = 225$.

Q22. Correct Answer: **12** • MEDIUM 🖋️ GRID-IN**STEP-BY-STEP SOLUTION:**

From the equation, $h = 9$ (since $x - 9$ matches $x - h$).

Rewrite $(y + 5)^2$ as $(y - (-5))^2$, so $k = -5$.

$r^2 = 64$, so $r = 8$.

$h + k + r = 9 + (-5) + 8 = 12$.

COMMON WRONG ANSWERS & WHY:

- Gridding 22 — using $k = 5$ instead of $k = -5$, missing the sign flip required when reading the center from $(y + 5)^2$.
- Gridding 76 — using $r^2 = 64$ in place of $r = 8$ in the final sum $(9 - 5 + 64)$.

TRAP ALERT 🚧

Getting each individual piece (h, k, r) almost right but losing a sign or forgetting to take the square root for r before combining everything in the final sum.

KEY INSIGHT 💡

When a question asks for a combination of h , k , and r , solve for each value completely and correctly first, then combine them as the very last step.

FAST METHOD ⚡

Identify $h = 9$, $k = -5$, $r = 8$ in three quick reads of the equation, then add: $9 - 5 + 8 = 12$.

Q23. Correct Answer: **B** • MEDIUM**STEP-BY-STEP SOLUTION:**

Isolate the constant: $(x^2 - 2x) + (y^2 - 14y) = -F$.

Complete the square: $x^2 - 2x$ becomes $(x - 1)^2 - 1$. $y^2 - 14y$ becomes $(y - 7)^2 - 49$.

Substitute: $(x - 1)^2 - 1 + (y - 7)^2 - 49 = -F$, so $(x - 1)^2 + (y - 7)^2 = -F + 50$.

Since radius is 9, $r^2 = 81$, so $-F + 50 = 81$.

Solve: $-F = 81 - 50 = 31$, so $F = -31$.

WHY EACH OPTION IS WRONG:

- A)** This drops the negative sign that results from solving $-F = 31$ for F .
- B)** Correct. $-F + 50 = 81$ leads to $-F = 31$, so $F = -31$.
- C)** This adds 50 and 81 together (getting 131) instead of subtracting to isolate F .
- D)** This combines both the addition error and the sign error, giving the negative of choice C.

TRAP ALERT 🚧

Setting up the equation $-F + 50 = 81$ correctly but then making a sign-tracking error while solving the final one-step equation for F .

KEY INSIGHT 💡

When solving for a constant like F that appears with a negative sign in the original equation, isolate it carefully and double-check the sign of your final answer.

FAST METHOD ⚡

The two completing-the-square constants always sum to the 'extra' amount ($1 + 49 = 50$ here). Set $50 - F = r^2$, so $F = 50 - r^2 = 50 - 81 = -31$.

Q24. Correct Answer: **C** • MEDIUM**STEP-BY-STEP SOLUTION:**

An x-intercept occurs where $y = 0$. Substitute $y = 0$ into the equation: $(x - 5)^2 + 0^2 = 9$.

Simplify: $(x - 5)^2 = 9$.

Take the square root of both sides: $x - 5 = \pm 3$.

Solve both cases: $x = 5 + 3 = 8$, or $x = 5 - 3 = 2$.

The x-intercepts are $(2, 0)$ and $(8, 0)$.

WHY EACH OPTION IS WRONG:

- A)** This confuses the center of the circle and the origin with the actual x-intercepts, which require solving the equation.
- B)** This applies a sign error, subtracting 3 from -5 instead of from the correct $h = 5$.
- C)** Correct. Setting $y = 0$ and solving gives $x = 5 \pm 3$, so $x = 2$ or $x = 8$.
- D)** This uses $\pm r = \pm 3$ directly as the x-intercepts, ignoring the horizontal shift caused by $h = 5$ entirely.

TRAP ALERT 🚧

Forgetting to add the center's x-coordinate back in after solving $x - 5 = \pm 3$, and instead reporting just ± 3 as if the circle were centered at the origin.

KEY INSIGHT 💡

To find intercepts of a circle, set the other variable to 0 and solve the resulting equation — don't assume the intercepts are simply $\pm r$ from the origin unless the circle is actually centered there.

FAST METHOD ⚡

Since $k = 0$ here, the circle's center sits directly on the x-axis, so the x-intercepts are simply $h - r$ and $h + r$: $5 - 3 = 2$ and $5 + 3 = 8$.

Q25. Correct Answer: **D** • MEDIUM**STEP-BY-STEP SOLUTION:**

The original center is $(2, -3)$, read from $(x - 2)^2 + (y + 3)^2$.

Translating 5 units right adds 5 to the x-coordinate: $2 + 5 = 7$. Translating 2 units up adds 2 to the y-coordinate: $-3 + 2 = -1$.

The new center is $(7, -1)$. The radius is unchanged, so r^2 is still 49.

The new equation is $(x - 7)^2 + (y + 1)^2 = 49$.

WHY EACH OPTION IS WRONG:

- A)** This moves the center in the opposite directions (left and down) instead of right and up, flipping both signs.
- B)** This applies the horizontal shift to the y-coordinate and the vertical shift to the x-coordinate, swapping which shift affects which axis.
- C)** This correctly finds the new center but mistakenly changes the radius, when a pure translation never changes a circle's size.

D) Correct. New center $(7, -1)$ keeps $r^2 = 49$ unchanged, since translating a circle never affects its radius.

TRAP ALERT 🚧

Assuming that shifting a circle also changes its radius, when in fact translation only moves the center — the size of the circle, and therefore r^2 , always stays exactly the same.

KEY INSIGHT 💡

Translating a circle shifts (h, k) by the given amounts but never changes r . Only the parentheses change; the number on the right side of the equation stays identical.

FAST METHOD ⚡

Add the horizontal shift to h and the vertical shift to k , then copy the original r^2 over unchanged — translation problems never require recalculating the radius.

Q26. Correct Answer: **18** • MEDIUM 🖋️ GRID-IN

STEP-BY-STEP SOLUTION:

Isolate the constant: $(x^2 + 12x) + (y^2 - 16y) = -19$.

Complete the square: $x^2 + 12x$ becomes $(x + 6)^2 - 36$. $y^2 - 16y$ becomes $(y - 8)^2 - 64$.

Substitute: $(x + 6)^2 - 36 + (y - 8)^2 - 64 = -19$, so $(x + 6)^2 + (y - 8)^2 = -19 + 36 + 64 = 81$.

So $r^2 = 81$, meaning $r = 9$.

The diameter is $2r = 2(9) = 18$.

COMMON WRONG ANSWERS & WHY:

- Gridding 9 — this is the radius, not the diameter. The question specifically asks for the diameter, which is twice this value.
- Gridding 81 — this is r^2 , the value immediately after completing the square, before taking the square root or doubling.

TRAP ALERT 🚧

Doing all the completing-the-square work correctly to find $r = 9$, then forgetting that the question asked for the diameter, not the radius.

KEY INSIGHT 💡

Always reread exactly what the question asks for at the end of a multi-step problem — radius, diameter, r^2 , and area are all easy to mix up after a long calculation.

FAST METHOD ⚡

Add the two completing-the-square constants and the isolated value ($36 + 64 - 19 = 81$), take the square root for radius (9), then double for diameter (18) in one continuous chain.

Q27. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

For the equation $x^2 + y^2 = k$, the right side equals r^2 .

Since $r = 7/2$, square this value: $r^2 = (7/2)^2 = 49/4$.

So $k = 49/4$.

WHY EACH OPTION IS WRONG:

A) Correct. $k = r^2 = (7/2)^2 = 49/4$.

- B) This uses r itself ($7/2$) directly as k , without squaring it.
- C) This squares only the numerator ($7^2 = 49$) while leaving the denominator unsquared, instead of correctly squaring the entire fraction.
- D) This doubles r ($7/2 \times 2 = 7$), treating it as if finding a diameter, then forgets to square at all.

TRAP ALERT 🚧

When squaring a fraction, squaring only the top number and forgetting that the bottom number must be squared too: $(7/2)^2 = 49/4$, not $49/2$.

KEY INSIGHT 💡

To square a fraction, square both the numerator and the denominator separately: $(a/b)^2 = a^2/b^2$.

FAST METHOD ⚡

Square numerator and denominator in one motion: $7^2 = 49$ and $2^2 = 4$, giving $49/4$ immediately.

Q28. Correct Answer: **B** • MEDIUM

STEP-BY-STEP SOLUTION:

The circumference of the circle is $C = 2\pi r$. We need $2\pi r > 40$.

Divide both sides by 2π : $r > 40/(2\pi) = 20/\pi$.

Since $\pi \approx 3.14159$, $20/\pi \approx 6.37$.

The integers from 1 to 10 that are greater than 6.37 are 7, 8, 9, and 10.

That is 4 values.

WHY EACH OPTION IS WRONG:

- A) This undercounts by one, possibly from a rounding error that excludes $r = 7$ incorrectly.
- B) Correct. Since r must exceed approximately 6.37, the qualifying integers are 7, 8, 9, and 10 — a total of 4 values.
- C) This overcounts by one, possibly from mistakenly including $r = 6$, which gives a circumference of about 37.7, not greater than 40.
- D) This includes both $r = 6$ and an extra value, doubly overcounting beyond the correct boundary.

TRAP ALERT 🚧

Approximating $20/\pi$ carelessly (for example, rounding π down to 3) and getting a boundary value that incorrectly includes or excludes $r = 6$ or $r = 7$.

KEY INSIGHT 💡

When an inequality involves π , use a precise enough approximation ($\pi \approx 3.14$) to correctly identify which integers satisfy the boundary, especially when the boundary value is close to a whole number.

FAST METHOD ⚡

Recall that π is just over 3, so $20/\pi$ is just under $20/3 \approx 6.67$, but more precisely about 6.37 — either way, 7 is clearly the smallest qualifying integer, giving 7 through 10, or 4 values.

Q29. Correct Answer: **C** • HARD

STEP-BY-STEP SOLUTION:

A circle tangent to the x -axis touches that axis at exactly one point, meaning the radius equals the perpendicular distance from the center to the x -axis.

The perpendicular distance from any point to the x-axis is simply the absolute value of its y-coordinate.

The center's y-coordinate is 6, so the radius is $|6| = 6$.

WHY EACH OPTION IS WRONG:

- A)** This uses the absolute value of the x-coordinate ($|-4| = 4$) instead of the y-coordinate, which is what matters for distance to the x-axis.
- B)** This adds the absolute values of both coordinates ($4 + 6 = 10$) instead of using only the y-coordinate.
- C)** Correct. Distance from $(-4, 6)$ to the x-axis is $|6| = 6$, which is the radius of a circle tangent to that axis.
- D)** This uses the x-coordinate directly without taking its absolute value, and the radius can never be negative regardless.

TRAP ALERT

Confusing tangency to the x-axis (which depends on the y-coordinate) with tangency to the y-axis (which depends on the x-coordinate), and using the wrong coordinate.

KEY INSIGHT

A circle tangent to the x-axis has radius equal to $|k|$ (the y-coordinate of the center); a circle tangent to the y-axis has radius equal to $|h|$ (the x-coordinate of the center).

FAST METHOD

Match the axis name to the opposite-sounding coordinate: tangent to the x-axis uses the y-coordinate's absolute value, and vice versa.

Q30. Correct Answer: **D** • **HARD**

STEP-BY-STEP SOLUTION:

Circle A has center $(1, 1)$ and radius $\sqrt{9} = 3$. Circle B has center $(9, 7)$ and radius $\sqrt{16} = 4$.

Find the distance between the centers: $d = \sqrt{(9 - 1)^2 + (7 - 1)^2} = \sqrt{64 + 36} = \sqrt{100} = 10$.

Compare this distance to the sum of the radii: $3 + 4 = 7$.

Since the distance between centers (10) is greater than the sum of the radii (7), the circles are too far apart to touch or overlap — each lies entirely outside the other.

WHY EACH OPTION IS WRONG:

- A)** Two intersection points require the distance between centers to be less than the sum of the radii but more than their difference; here the distance (10) exceeds even the sum (7).
- B)** External tangency requires the distance between centers to equal exactly the sum of the radii (7); here the distance is 10, not 7.
- C)** One circle being inside another requires the distance between centers to be less than the difference of the radii ($|4 - 3| = 1$); here the distance (10) is far larger.
- D)** Correct. Since $10 > 3 + 4$, the circles are separated with no overlap — each lies entirely outside the other.

TRAP ALERT

Not knowing the three key distance benchmarks (sum of radii, difference of radii) that determine whether two circles intersect, are tangent, or sit entirely apart.

KEY INSIGHT

Comparing the distance between two centers to the sum and difference of their radii tells you everything: distance $>$ sum means separate; distance $=$ sum means externally tangent; distance

between the difference and the sum means two intersection points; distance < difference means one circle is inside the other.

FAST METHOD ⚡

Just compute distance and sum of radii first — if distance clearly exceeds the sum, the circles are separate with no further checking needed.

Q31. Correct Answer: **7** • **HARD** 🖊 **GRID-IN**

STEP-BY-STEP SOLUTION:

Isolate the constant: $(x^2 - 22x) + (y^2 + 10y) = -97$.

Complete the square: $x^2 - 22x$ becomes $(x - 11)^2 - 121$. $y^2 + 10y$ becomes $(y + 5)^2 - 25$.

Substitute: $(x - 11)^2 - 121 + (y + 5)^2 - 25 = -97$, so $(x - 11)^2 + (y + 5)^2 = -97 + 121 + 25 = 49$.

So $r^2 = 49$, which means $r = 7$.

COMMON WRONG ANSWERS & WHY:

- Gridding 49 — this is r^2 , not r . With larger coefficients like -22 , it's especially easy to stop right after completing the square without taking the final square root.
- Gridding 11 — mistakenly gridding the x -coordinate of the center ($h = 11$) instead of the radius.

TRAP ALERT 🚧

With bigger numbers like -22 and 10 , the completing-the-square arithmetic takes more steps, increasing the chance of stopping early at $r^2 = 49$ or making an arithmetic slip with the larger constants 121 and 25 .

KEY INSIGHT 💡

The size of the coefficients doesn't change the method — isolate the constant, complete the square on each variable, then take a final square root. Larger numbers just call for extra care.

FAST METHOD ⚡

Half of -22 is -11 , squared is 121 . Half of 10 is 5 , squared is 25 . Add both to the isolated value: $121 + 25 - 97 = 49$, then $\sqrt{49} = 7$.

Q32. Correct Answer: **B** • **HARD**

STEP-BY-STEP SOLUTION:

A circle tangent to the y -axis has a radius equal to the perpendicular distance from its center to the y -axis, which is $|h|$, the absolute value of the x -coordinate.

So $|h| = 13$, meaning $h = 13$ or $h = -13$.

Since the center lies in the second quadrant, its x -coordinate must be negative (second-quadrant points have negative x and positive y).

Therefore $h = -13$.

WHY EACH OPTION IS WRONG:

- A)** This correctly finds $|h| = 13$ but ignores the quadrant condition, which requires h to be negative.
- B)** Correct. $|h| = 13$, and since the center is in the second quadrant (negative x , positive y), $h = -13$.
- C)** This mistakenly uses the given y -coordinate (4) instead of solving for h using the tangency condition.

D) This applies the correct negative sign but to the wrong coordinate value (4 instead of 13).

TRAP ALERT 🚧

Correctly setting up $|h| = 13$ from the tangency condition but then forgetting to apply the quadrant information that determines which sign is correct.

KEY INSIGHT 💡

Tangency conditions give you the absolute value of a coordinate; additional information like a stated quadrant is what tells you the correct sign to use.

FAST METHOD ⚡

Translate quadrant descriptions immediately into sign rules: Quadrant II is $(-, +)$, so as soon as you find $|h| = 13$, attach a negative sign.

Q33. Correct Answer: **B • HARD**

STEP-BY-STEP SOLUTION:

Tangency to the horizontal line $y = 11$ means the radius equals the perpendicular distance from the center to that line.

The center is $(-2, 5)$. The perpendicular distance from $(-2, 5)$ to the line $y = 11$ is $|11 - 5| = 6$.

So $r = 6$, which means $r^2 = 36$.

WHY EACH OPTION IS WRONG:

A) This is r , not r^2 — the question specifically asks for the squared value.

B) Correct. The distance from the center to the line $y = 11$ is 6, so $r = 6$ and $r^2 = 36$.

C) This squares the y -coordinate of the line itself ($11^2 = 121$) instead of squaring the actual distance from center to line.

D) This squares the y -coordinate of the center ($5^2 = 25$) instead of computing and squaring the distance to the line.

TRAP ALERT 🚧

Confusing the numbers given in the problem (the line's y -value or the center's y -coordinate) with the distance that actually needs to be calculated and then squared.

KEY INSIGHT 💡

For a circle tangent to a horizontal line $y = n$, the radius equals the vertical distance from the center's y -coordinate to n ; for a vertical line $x = n$, use the horizontal distance instead.

FAST METHOD ⚡

Subtract the center's relevant coordinate from the line's value, take the absolute value for r , then square it: $|11 - 5| = 6$, so $r^2 = 36$.

Q34. Correct Answer: **C • HARD**

STEP-BY-STEP SOLUTION:

Before completing the square, the coefficients on x^2 and y^2 must both be 1. Divide the entire equation by 2: $x^2 + y^2 - 4x + 12y + 15 = 0$.

Isolate the constant: $(x^2 - 4x) + (y^2 + 12y) = -15$.

Complete the square: $x^2 - 4x$ becomes $(x - 2)^2 - 4$. $y^2 + 12y$ becomes $(y + 6)^2 - 36$.

Substitute: $(x - 2)^2 - 4 + (y + 6)^2 - 36 = -15$, so $(x - 2)^2 + (y + 6)^2 = -15 + 4 + 36 = 25$.

So $r^2 = 25$, which means $r = 5$.

WHY EACH OPTION IS WRONG:

- A)** This is r^2 , not r — the final square root step was skipped after correctly dividing by 2 and completing the square.
- B)** This comes from forgetting to divide the equation by 2 before completing the square, roughly doubling the resulting radius.
- C)** Correct. After dividing by 2 to make the leading coefficients 1, completing the square gives $r^2 = 25$, so $r = 5$.
- D)** This divides the already-correct radius by 2 a second, unnecessary time.

TRAP ALERT

Jumping straight into completing the square while the x^2 and y^2 terms still have a coefficient of 2, which makes every subsequent step invalid and inflates the final radius.

KEY INSIGHT

Standard form completing-the-square only works when the coefficients on x^2 and y^2 are exactly 1. Always divide the entire equation by that shared coefficient first.

FAST METHOD

Check the coefficient on x^2 and y^2 before doing anything else — if it isn't 1, divide every single term by it immediately, including the constant.

Q35. Correct Answer: **21** • **HARD**  **GRID-IN**

STEP-BY-STEP SOLUTION:

Since the coefficients on x^2 and y^2 are both 3, divide the entire equation by 3 first: $x^2 + y^2 + 10x - 6y + 13 = 0$.

Isolate the constant: $(x^2 + 10x) + (y^2 - 6y) = -13$.

Complete the square: $x^2 + 10x$ becomes $(x + 5)^2 - 25$. $y^2 - 6y$ becomes $(y - 3)^2 - 9$.

Substitute: $(x + 5)^2 - 25 + (y - 3)^2 - 9 = -13$, so $(x + 5)^2 + (y - 3)^2 = -13 + 25 + 9 = 21$.

So $r^2 = 21$.

COMMON WRONG ANSWERS & WHY:

- Gridding 63 — forgetting to divide the constant term by 3 along with every other term, using the original 39 in place of 13.
- Gridding 4.58 (approximately) — taking an unnecessary square root, since the question specifically asks for r^2 , not r .

TRAP ALERT

Remembering to divide the x^2 , y^2 , x , and y coefficients by 3 but forgetting that the constant term must be divided by 3 as well, since every single term in the equation needs the same treatment.

KEY INSIGHT

When dividing an equation to make the leading coefficients 1, every term — including the constant — must be divided by the same number, with no exceptions.

FAST METHOD

Divide the whole equation by 3 in one pass before writing anything else down, so every later step works with already-simplified numbers (10, -6, 13).

Q36. Correct Answer: **D** • **HARD**

STEP-BY-STEP SOLUTION:

Isolate the constant: $(x^2 - 14x) + (y^2 + 6y) = -k$.

Complete the square: $x^2 - 14x$ becomes $(x - 7)^2 - 49$. $y^2 + 6y$ becomes $(y + 3)^2 - 9$.

Substitute: $(x - 7)^2 - 49 + (y + 3)^2 - 9 = -k$, so $(x - 7)^2 + (y + 3)^2 = -k + 58$.

For the equation to represent exactly one point (a circle with radius 0), the right side must equal 0:
 $-k + 58 = 0$.

Solve: $k = 58$.

WHY EACH OPTION IS WRONG:

- A)** This assumes k itself should be 0, rather than recognizing that the right side of the completed-square equation (r^2) is what must equal 0.
- B)** This solves $-k + 58 = 0$ with a sign error, getting the negative of the correct answer.
- C)** This incorrectly halves 58 instead of solving the equation $-k + 58 = 0$ directly.
- D)** Correct. Setting $r^2 = -k + 58$ equal to 0 gives $k = 58$, which makes the 'circle' shrink to a single point.

TRAP ALERT

Misunderstanding what 'exactly one point' means algebraically — it requires the right-hand side of the completed-square equation (r^2) to equal exactly 0, not the constant k itself.

KEY INSIGHT

A circle equation with $r^2 = 0$ represents a single point (the center itself), since the only solution is the point at distance 0 from the center. If r^2 were negative, no real points would satisfy the equation at all.

FAST METHOD

Complete the square as usual to find the 'extra' constant ($49 + 9 = 58$ here), then simply set that equal to k for the single-point case, since that makes the right side become exactly 0.

Q37. Correct Answer: **A** • HARD

STEP-BY-STEP SOLUTION:

Since $\text{area} = \pi r^2 = 144\pi$, divide both sides by π to get $r^2 = 144$, so $r = 12$.

Find the distance from the garden's center (20, 35) to the sprinkler head (20, 23). Since both points share the same x -coordinate, the distance is the difference in y -coordinates: $|35 - 23| = 12$.

Since this distance (12) exactly equals the radius (12), the sprinkler head sits exactly on the boundary of the garden.

WHY EACH OPTION IS WRONG:

- A)** Correct. The distance from center to sprinkler head is exactly 12, which equals the radius, placing the point precisely on the boundary.
- B)** This would require the distance to be less than 12, but the distance is exactly 12, not smaller.
- C)** This would require the distance to be greater than 12, but the distance is exactly 12, not larger.
- D)** The position can absolutely be determined: the area gives the radius, and the two points' coordinates give the distance directly.

TRAP ALERT

Computing the radius and the distance correctly but then assuming any computed numbers being equal must be a coincidence rather than the actual mathematical condition for lying exactly on the boundary.

KEY INSIGHT

A point lies on a circle exactly when its distance from the center equals the radius — no more, no less. This is the same comparison used for general inside/on/outside questions, just applied to a real-world context.

FAST METHOD ⚡

Convert area to r^2 first (divide by π), take the square root for r , then compare directly to the distance between the two given points — here, both equal 12 immediately.

Q38. Correct Answer: **B** • **HARD**

STEP-BY-STEP SOLUTION:

The circle has center $(3, -2)$ and radius $\sqrt{100} = 10$.

Find the perpendicular distance from the center to the chord's line, $y = 6$: $d = |6 - (-2)| = 8$.

The radius, the perpendicular distance to the chord, and half the chord form a right triangle, so half the chord length $= \sqrt{(r^2 - d^2)} = \sqrt{(100 - 64)} = \sqrt{36} = 6$.

The full chord length is twice this value: $2 \times 6 = 12$.

WHY EACH OPTION IS WRONG:

A) This is half the chord length — the calculation correctly finds 6 but stops before doubling it for the full chord.

B) Correct. Half-chord $= \sqrt{(100 - 64)} = 6$, so the full chord length is $2(6) = 12$.

C) This adds the radius and the perpendicular distance pieces incorrectly instead of using the Pythagorean relationship between them.

D) This uses the diameter (20) directly as if the chord passed through the center, ignoring that the chord lies 8 units away from the center.

TRAP ALERT 🚨

Forgetting that the segment connecting the center perpendicular to a chord bisects that chord, so the right-triangle calculation only gives HALF the chord length, requiring one final doubling step.

KEY INSIGHT 💡

For any chord, the radius, the perpendicular distance from center to chord, and half the chord length always form a right triangle: $r^2 = d^2 + (\text{half-chord})^2$.

FAST METHOD ⚡

Recognize the right triangle relationship immediately: with $r = 10$ and $d = 8$, this is a 6-8-10 triangle, so half the chord is 6 and the full chord is 12.

Q39. Correct Answer: **24** • **HARD** 🖋️ **GRID-IN**

STEP-BY-STEP SOLUTION:

The circle has center $(-4, 1)$ and radius $\sqrt{169} = 13$.

Find the perpendicular distance from the center to the chord's line, $x = 1$: $d = |1 - (-4)| = 5$.

Using the right-triangle relationship $r^2 = d^2 + (\text{half-chord})^2$: half-chord $= \sqrt{(169 - 25)} = \sqrt{144} = 12$.

The full chord length is $2 \times 12 = 24$.

COMMON WRONG ANSWERS & WHY:

- Gridding 12 — this is only half the chord length; the final doubling step was skipped.
- Gridding 26 — this is the diameter (2×13), used by mistake as if the chord passed directly through the center, ignoring the 5-unit offset.

TRAP ALERT 🚧

Stopping at the half-chord value (12) since it's the direct output of the right-triangle calculation, without remembering to double it for the actual chord length asked for.

KEY INSIGHT 💡

Whenever a chord doesn't pass through the center, set up the right triangle $r^2 = d^2 + (\text{half-chord})^2$ using the perpendicular distance from center to the chord's line, then double the result for the full chord.

FAST METHOD ⚡

Spot the 5-12-13 Pythagorean triple immediately from $r = 13$ and $d = 5$, giving half-chord = 12 and full chord = 24 with no decimal computation needed.

Q40. Correct Answer: **C** • **HARD****STEP-BY-STEP SOLUTION:**

From the larger circle's equation, $r^2 = 169$, so its area is $\pi(169) = 169\pi$.

The smaller circle has radius 5, so its area is $\pi(5^2) = 25\pi$.

The area between the two circles is the larger area minus the smaller area: $169\pi - 25\pi = 144\pi$.

WHY EACH OPTION IS WRONG:

- A)** This uses only the larger circle's area, forgetting to subtract the smaller circle's area entirely.
- B)** This comes from subtracting the radii first and then squaring the result, $(13 - 5)^2 = 64$, instead of subtracting the two areas.
- C)** Correct. Subtracting the areas directly: $169\pi - 25\pi = 144\pi$.
- D)** This adds the two areas together ($169\pi + 25\pi = 194\pi$) instead of subtracting to find the region between them.

TRAP ALERT 🚧

Subtracting the radii ($13 - 5 = 8$) before squaring, instead of squaring each radius separately first and then subtracting the resulting areas — these give very different results because squaring does not distribute over subtraction.

KEY INSIGHT 💡

Areas don't subtract the way radii do: $(a - b)^2$ is never equal to $a^2 - b^2$. Always compute each full area first, then subtract.

FAST METHOD ⚡

Find each r^2 separately (169 and 25), subtract those (144), and only then attach π — never subtract the radii themselves before squaring.

TOPIC SUMMARY — Circle Equations**Common Mistakes:**

1. Misreading the sign of a center coordinate: $(x - h)^2 + (y - k)^2 = r^2$ always uses the OPPOSITE sign of what appears inside the parentheses, so $(x + 3)^2$ means $h = -3$, not 3.
2. Confusing r and r^2 : the constant on the right side of standard form is always r^2 . Many questions ask for the radius, diameter, area, or circumference, each requiring a different next step after finding r^2 .

3. Forgetting to divide by the leading coefficient before completing the square when the equation is written as $Ax^2 + Ay^2 + \dots = 0$ with $A \neq 1$, or forgetting to move the constant to the correct side before completing the square at all.

Must Remember:

1. Standard form: $(x - h)^2 + (y - k)^2 = r^2$, with center (h, k) and radius r . Diameter $= 2r$. Circumference $= 2\pi r$. Area $= \pi r^2$.
2. To convert general form $x^2 + y^2 + Dx + Ey + F = 0$ to standard form: isolate the constant, complete the square on the x -terms and y -terms separately (using half the coefficient, squared), then add those values to both sides.
3. Key relationships: a circle tangent to the x -axis has $r = |k|$; tangent to the y -axis has $r = |h|$; for a chord not through the center, $r^2 = d^2 + (\text{half-chord})^2$, where d is the perpendicular distance from center to chord.

Expected on DSAT:

Based on this project's frequency analysis of past Digital SAT papers, circle equations appeared in 11 of 1,240 math questions analyzed (about 0.9%). That works out to roughly 0–1 question per test on average, with the topic showing up in only about 1 of every 3 tests — but when it does appear, expect a direct test of standard form, completing the square, or center/radius identification rather than a multi-part word problem.

— END OF SOLUTIONS —